

ROBOCUPJUNIOR RESCUE MAZE 2020
TEAM DESCRIPTION PAPER - TEAM simple

Mechanical Design and Manufacturing

Lightweight construction, articulated mobility, and reliable rescue-kit deployment

3.a Mechanical Design and Manufacturing

The robot was designed to move quickly and precisely through a maze while remaining capable of crossing ramps, steps, and uneven terrain. The mechanical structure combines lightweight LEGO components with purpose-built 3D-printed modules. The resulting design reduces unnecessary mass, maintains wheel contact on difficult surfaces, protects sensitive electronics, and provides controlled bidirectional rescue-kit deployment.



Figure M1. Overall mechanical configuration of the rescue-maze robot.

1. Main Structure

The core frame is constructed primarily from LEGO components. Compared with heavy acrylic or metal plates, the lightweight frame reduces the load on the drive motors and limits forward momentum during braking. This helps the robot stop more accurately at tile boundaries and reduces skidding during direction changes. Custom 3D-printed supports are used only where standard LEGO parts cannot provide the required geometry or reinforcement.

2. Actuators and Power Train

The drive system uses four large LEGO motors and large rubber wheels. The front and rear chassis sections are connected through a central articulated joint so that they can rotate relative to each other. When the robot climbs a ramp or crosses a step, this joint allows the chassis to conform to the terrain instead of forcing the wheels to remain on one rigid plane. This keeps all four wheels in contact with the surface, improves traction, and reduces wheel spin. The large wheel diameter also lowers the risk of the underside becoming trapped on an obstacle edge.

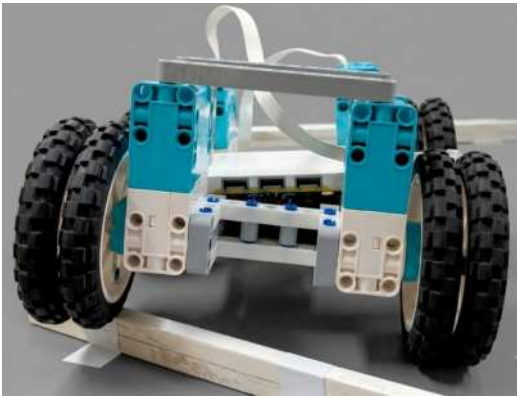


Figure M2. Articulated chassis structure and central pivot.

3. Subassemblies and Modules

Camera Mount Module

The camera mount holds the OpenMV camera securely while preserving a wide viewing angle. It is positioned as far inward as possible to protect the lens and connector from contact with maze walls. The mount is symmetrical to improve alignment and includes cable support to prevent the connector from loosening during movement. Because the wall height is 15 cm, the camera is placed near the mid-height of the wall, approximately 7.5 cm above the floor.

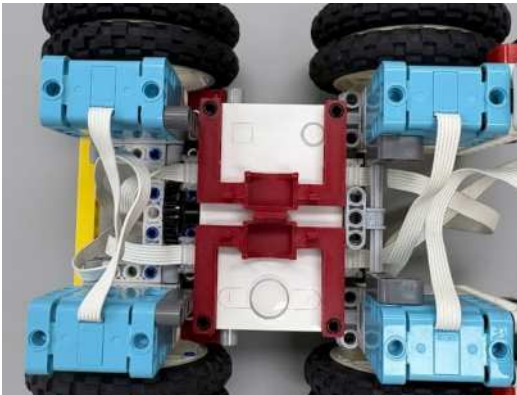


Figure M3. Camera mount geometry and protected lens position.

Rescue-Kit Storage and Deployment Module

The rescue-kit module combines a storage container, left and right deployment ramps, and a release mechanism. The container can hold up to eight rescue kits. The ramps guide each kit toward the selected side of the robot, and the release mechanism dispenses kits individually. This configuration enables accurate placement without rotating the entire robot toward the victim.



Figure M4. 3D-printed bidirectional rescue-kit ramp.

Distance-Sensor Mount Module

Three distance sensors are mounted at the front and sides of the robot. Their positions provide clear forward, left, and right measurement directions while keeping the sensors mechanically protected. Placing the sensors near the front of the robot increases the available reaction distance before the chassis reaches a wall or obstacle.

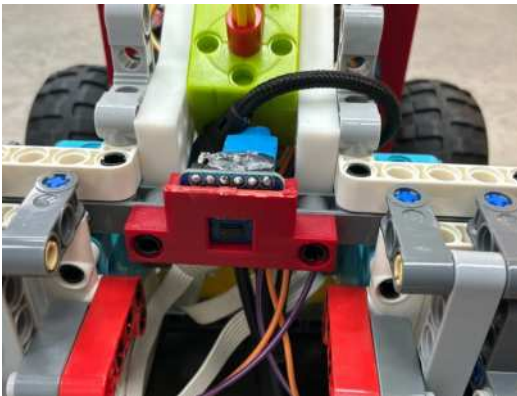


Figure M5. Front and side distance-sensor mounts.

Handle and Structural Reinforcement Module

A dedicated handle allows the robot to be lifted and transported without pulling on cables or sensitive components. The surrounding reinforcement links the handle to the main chassis and helps prevent separation between the chassis sections and the central turntable during repeated obstacle-crossing runs.

4. Rescue-Kit Deployment Mechanism

The bidirectional deployment mechanism was custom-built with 3D printing. Its slide-shaped ramps allow each rescue kit to move downward under its own weight, reducing the need for complex mechanical parts. A small control motor operates the upper release mechanism and selects the left or right deployment direction. Because the robot does not need to rotate its entire body before delivering a kit, the design reduces mission time and simplifies victim servicing in narrow passages.

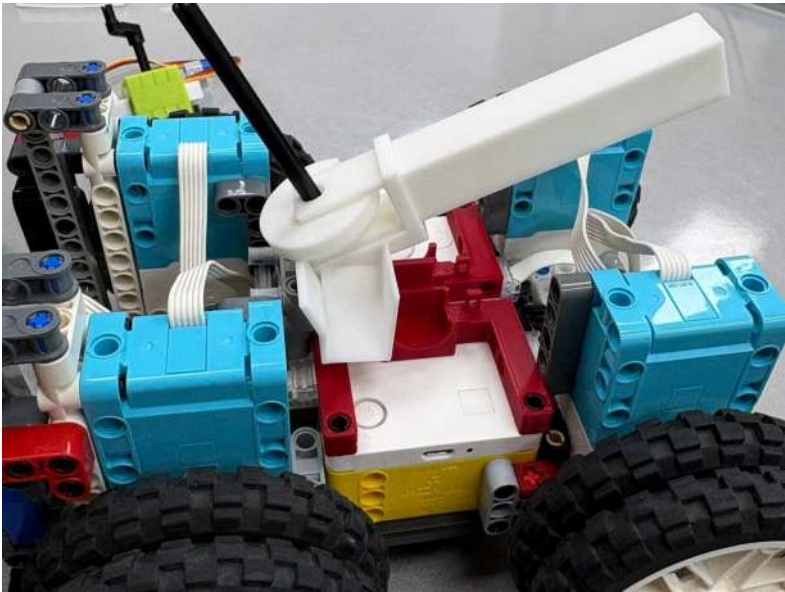


Figure M6. Controlled left-right rescue-kit release structure.

During iterative assembly, the ramp angle, opening size, and release speed were adjusted to prevent a kit from catching on the printed structure. The final design emphasizes gravity-assisted motion, a short release path, and a mechanically simple gate. These choices improve repeatability while keeping the mechanism compact.

5. Mechanical Validation and Iteration

Validation Item	Purpose	Design Response
Braking and turning	Check whether the robot stops accurately at tile boundaries and avoids excessive skidding.	Reduced mass and large rubber wheels improve stopping control and grip.
Ramp and step crossing	Confirm that the wheels maintain contact with uneven surfaces.	The articulated chassis allows front and rear sections to adapt independently.
Camera stability	Check lens alignment and cable security during repeated movement.	The inward-mounted, reinforced camera holder protects the lens and connector.
Kit deployment	Verify one-by-one release to either side without rotating the chassis.	The gravity-assisted ramp and controlled gate reduce jamming risk.

ROBOCUPJUNIOR RESCUE MAZE 2020
TEAM DESCRIPTION PAPER - TEAM simple
Electronic Design and Manufacturing
Controller, sensor network, power distribution, and actuator interfaces

3.b Electronic Design and Manufacturing

The electronic system is divided between the LEGO SPIKE Prime Hub and the OpenMV H7 vision-and-sensor subsystem. The SPIKE Prime Hub is responsible for drive control, IMU-based motion correction, and high-level navigation. The OpenMV H7 processes visual targets, reads three VL53L4CD time-of-flight distance sensors and two touch sensors, and receives commands for the rescue-kit release motor. A PUPRemote interface links the two devices so that sensor data and deployment commands can be exchanged during autonomous operation.

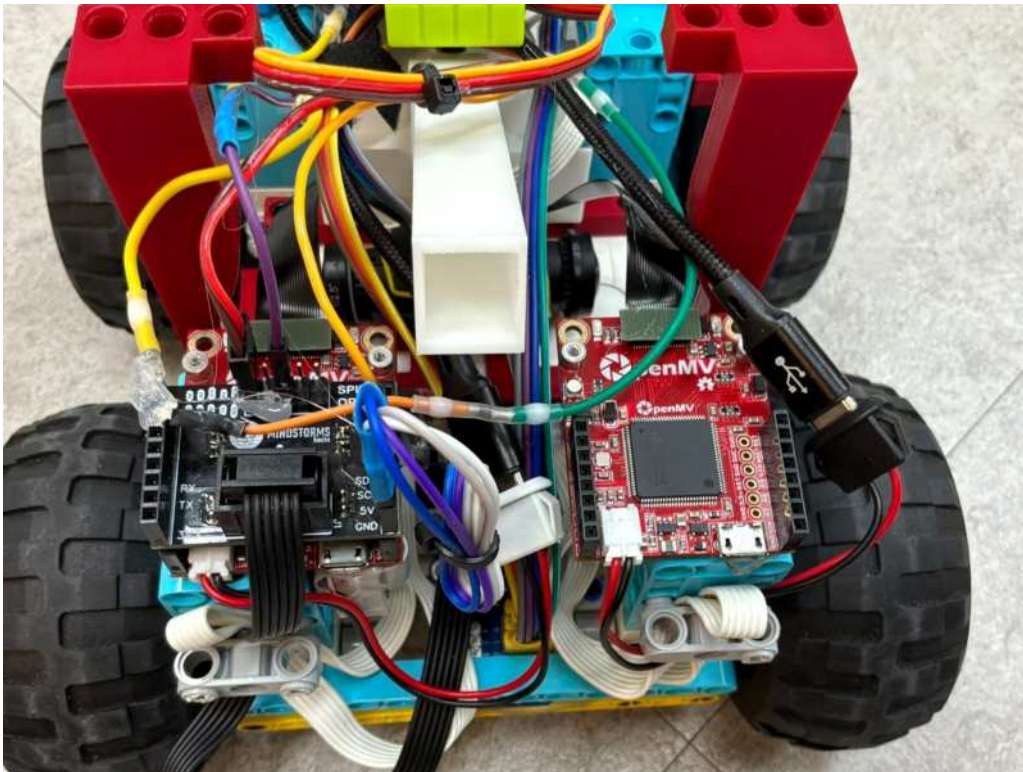


Figure E1. Top view of the controller, OpenMV H7 board, wiring, and interface modules.

1. Main Controller and Actuators

The LEGO SPIKE Prime Hub serves as the main controller. Four LEGO drive motors provide the traction required to move the large off-road wheels and cross ramps or steps. The hub also uses its built-in IMU to maintain the desired heading and reduce drift during straight driving and turning. A separate small motor controls the rescue-kit release mechanism so that kits can be dispensed accurately in the required direction.

Electronic Component	Primary Role
LEGO SPIKE Prime Hub	Main navigation controller, motor control, coordinate mapping, and built-in IMU processing.
Four LEGO drive motors	Independent traction for the four large wheels and stable obstacle crossing.
Small deployment motor	Controls the rescue-kit release gate and the selected deployment direction.
OpenMV H7	Vision processing, distance-sensor acquisition, touch-sensor input, and communication with the hub.

2. Sensors Used - Vision, IMU, Distance, and Touch

Vision Sensor: One OpenMV H7 camera is mounted inside the chassis with its lens directed toward the wall area. The camera identifies concentric cognitive targets and the Greek-letter victims Φ , Ψ , and Ω . The protected inward position limits mechanical interference while maintaining a useful field of view.

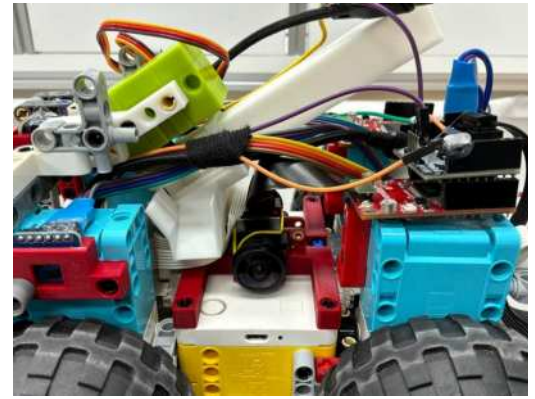


Figure E2. OpenMV H7 board and protected camera position.

Distance Sensors: Three VL53L4CD time-of-flight sensors measure the distance to the left, front, and right sides of the robot. These readings support wall detection, wall-following correction, and collision avoidance. The sensor mounts are aligned to preserve clear measurement directions near the front of the chassis.

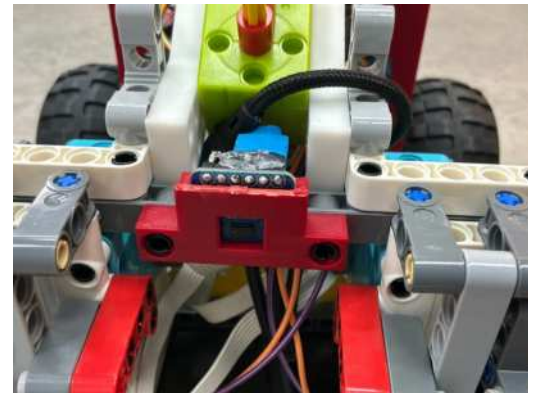


Figure E3. Forward and side distance-sensor arrangement.

IMU and Touch Sensors: The built-in SPIKE Prime IMU is used for heading correction and turn control. Two touch sensors provide a mechanical backup for close-range contact and wall alignment. If a distance reading is insufficient during a close approach, the touch sensors allow the robot to stop and correct its position.

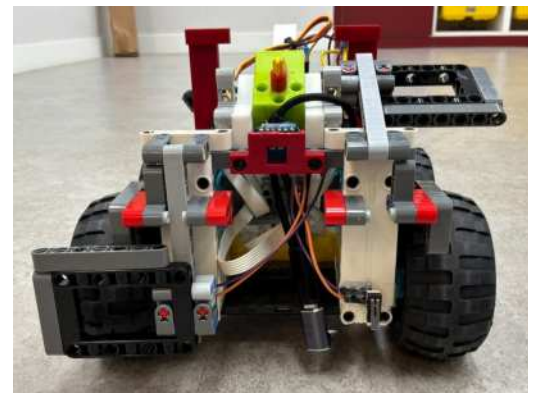


Figure E4. Front structure containing the protected contact-sensing area.

3. Power Subsystem

The robot uses the SPIKE Prime Hub battery together with an auxiliary external battery pack for the additional electronics. Separating the main controller supply from the external vision-and-sensor load helps maintain stable operation during longer missions and during high-load actions such as ramp climbing, obstacle crossing, and rescue-kit deployment. The cable routing is fixed to the chassis to reduce connector strain during repeated movement.

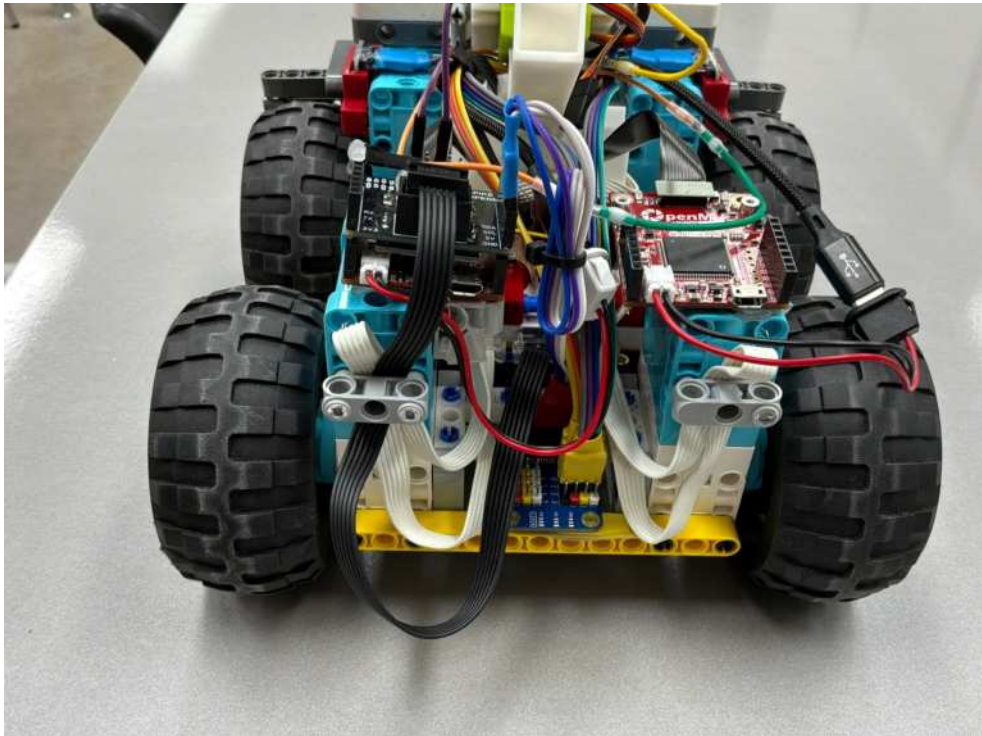


Figure E5. Rear view of the electronic subsystem, interface board, and cable routing.

4. Communication and Integration

PUPRemote is used to connect the OpenMV-based subsystem to the SPIKE Prime Hub. OpenMV operates as an external Powered Up sensor. It sends the three distance readings, touch-sensor states, and victim-detection results to the hub. After the hub determines the required rescue-kit quantity and direction, it returns a deployment command to OpenMV. This bidirectional structure keeps navigation decisions in the main controller while allowing the vision subsystem to continue camera processing and sensor acquisition.

Data Flow	Transferred Information
OpenMV H7 -> SPIKE Prime Hub	Left/front/right ToF distances, two touch-sensor states, cognitive-target score, and letter-victim result.
SPIKE Prime Hub -> OpenMV H7	Rescue-kit quantity and deployment-direction command.
OpenMV H7 -> Deployment Motor	Queued motor operation for controlled kit release while sensing continues.

5. Electronic Validation and Iteration

Validation Item	Verification Focus	Design Response
Distance sensing	Confirm stable left/front/right wall measurements during movement.	Rigid front-and-side mounting maintains consistent sensor alignment.
Vision processing	Check victim recognition while the robot is not perfectly centered.	The camera is protected and positioned to preserve a wide usable view.
Contact backup	Verify safe response during close wall approaches.	Two touch sensors provide a mechanical fallback for stopping and correction.
Power stability	Check operation during driving, obstacle crossing, and kit release.	The auxiliary battery pack supports external electronics and reduces supply fluctuation.
Communication	Confirm that sensing continues while deployment commands are executed.	Bidirectional PUPRemote communication separates data acquisition from actuation commands.